Theory Concepts - Data Preprocessing & Feature Engineering Project

1. What is Data Analysis?

Data analysis is the process of collecting, cleaning, transforming, and interpreting data to discover useful information, identify patterns, and support decision-making.

2. Planning a Data Science Project

Steps include: Problem Definition, Data Collection, Data Understanding, Data Cleaning and Preprocessing, Exploratory Data Analysis, Feature Engineering, Model Building, Model Evaluation, Deployment, and Monitoring.

3. Framing a Machine Learning Problem

Identify the business problem, define the target variable, select input features, determine whether the task is classification or regression, choose evaluation metrics, and prepare the data for machine learning.

4. Tensors with NumPy Examples

A tensor is a multidimensional array. Examples: Scalar (0D): np.array(10); Vector (1D): np.array([1,2,3]); Matrix (2D): np.array([[1,2],[3,4]]); 3D Tensor: np.array([[[1,2],[3,4]], [[5,6],[7,8]]]). Tensors are the fundamental data structures used in machine learning and deep learning.